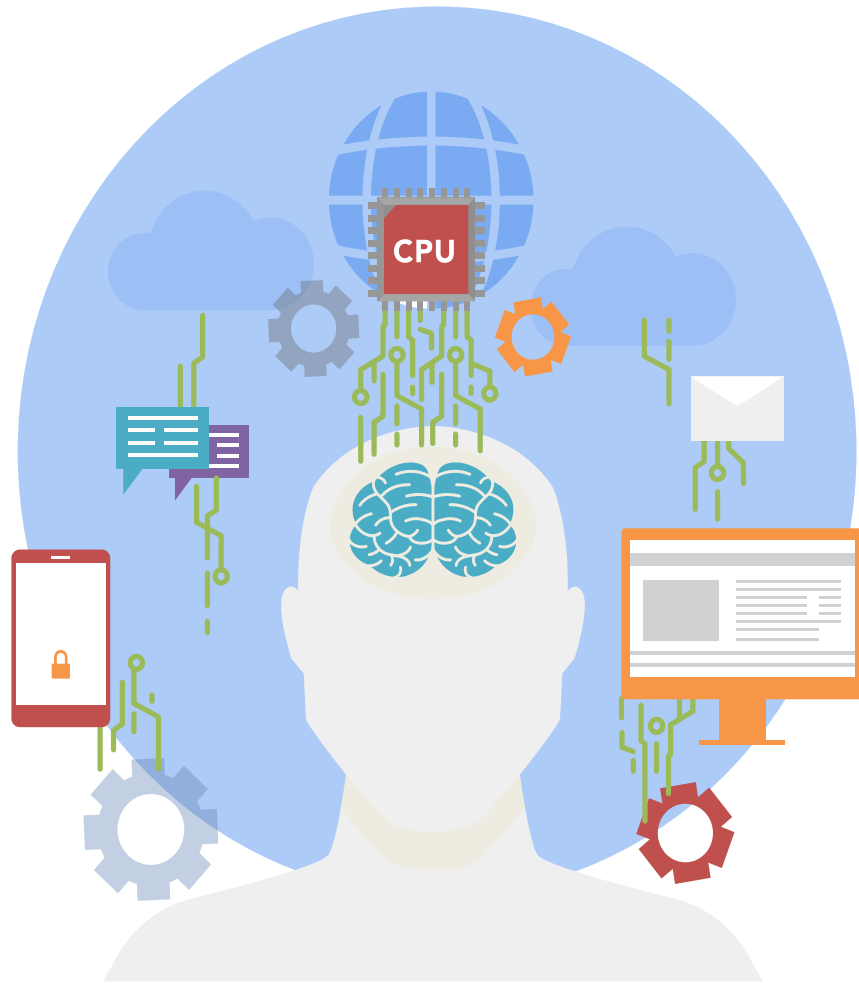




Understanding Agentic AI

Dr. Tamer Arafa



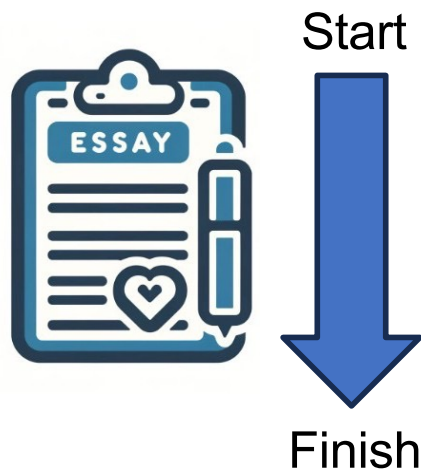
Introduction to Agentic Workflows

What is Agentic AI?

Agentic AI

Non-agentic workflow (zero-shot):

Please type out an essay on topic X from start to finish in one go, without using backspace.



Agentic workflow:

Write an essay outline on topic X

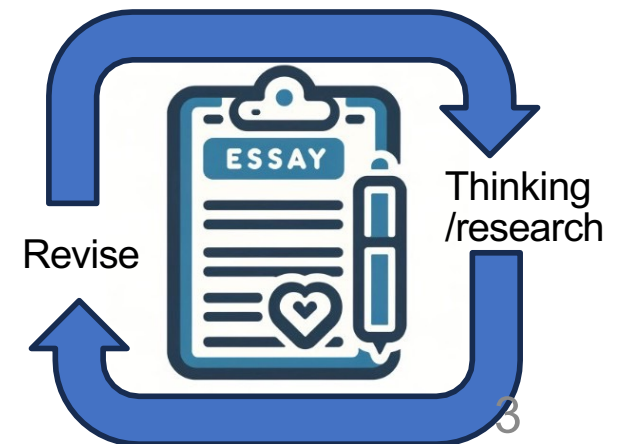
Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.

...



Agentic AI workflows

An agentic AI workflow is a process where an LLM-based app executes multiple steps to complete a task.

Essay-writing example:

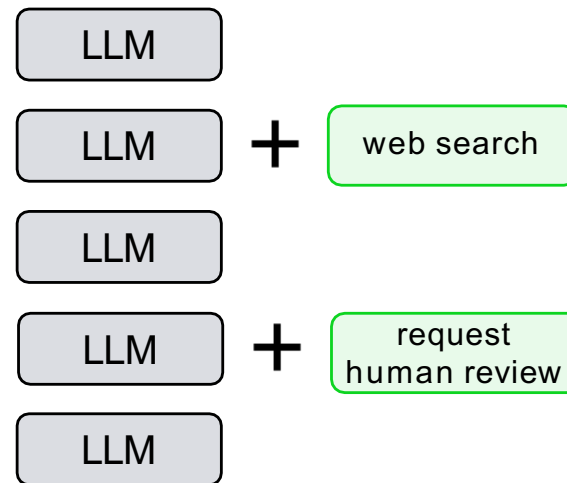
Write an essay outline on topic X

Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.





Sequoia Ascent, March 2024



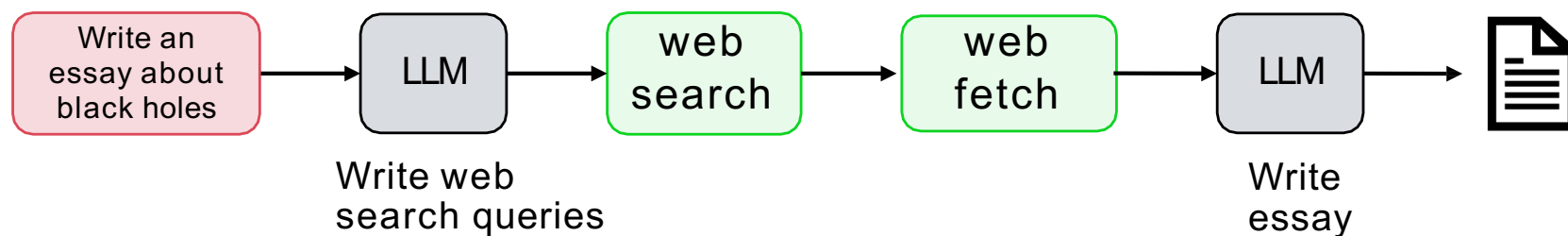
....
Rather than arguing over which work to include or exclude as being a true agent, we can acknowledge that there are different degrees to which systems can be agentic.
....



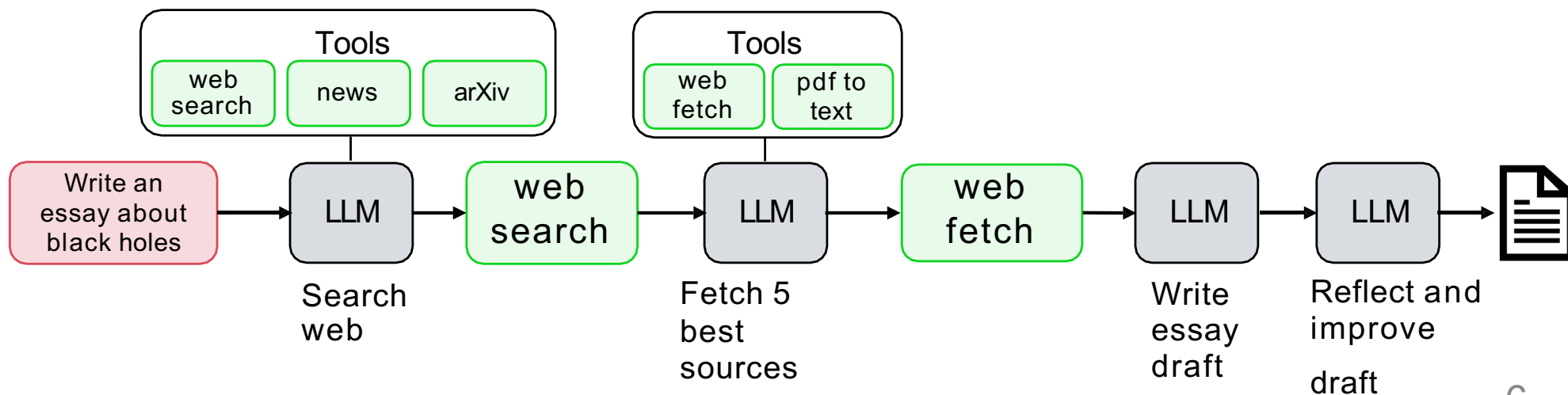
X (twitter) post , June 2024

Degrees of autonomy

Less autonomous

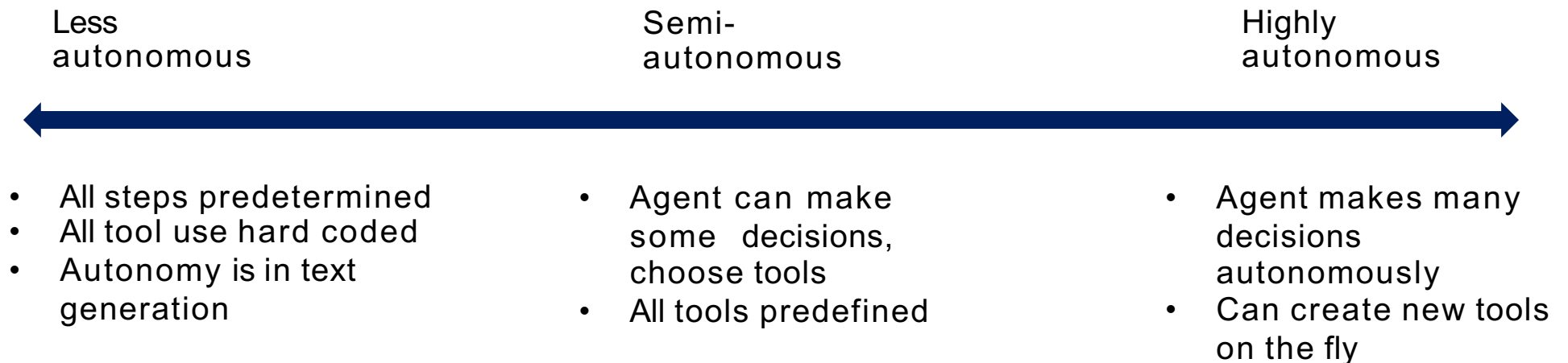


More autonomous

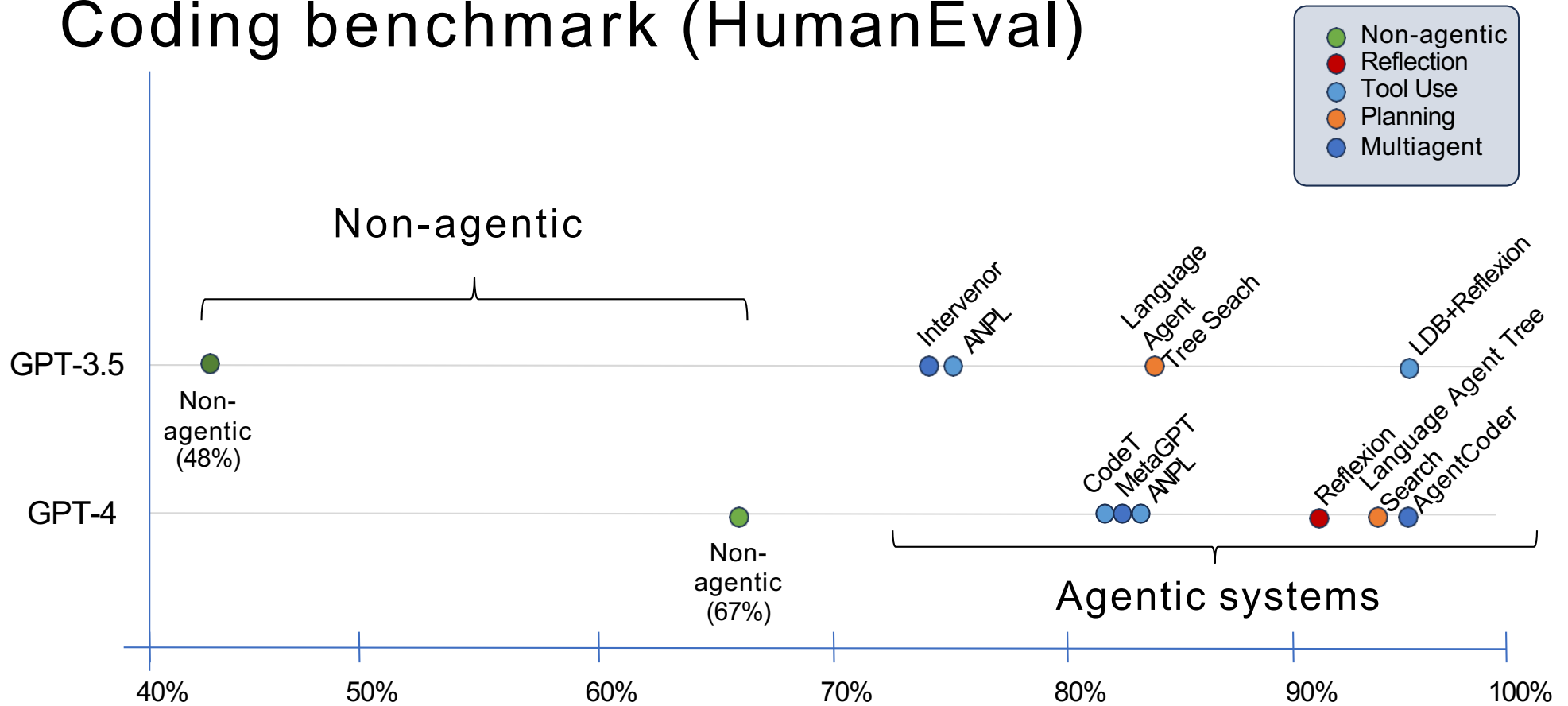


Degrees of “agenticness”

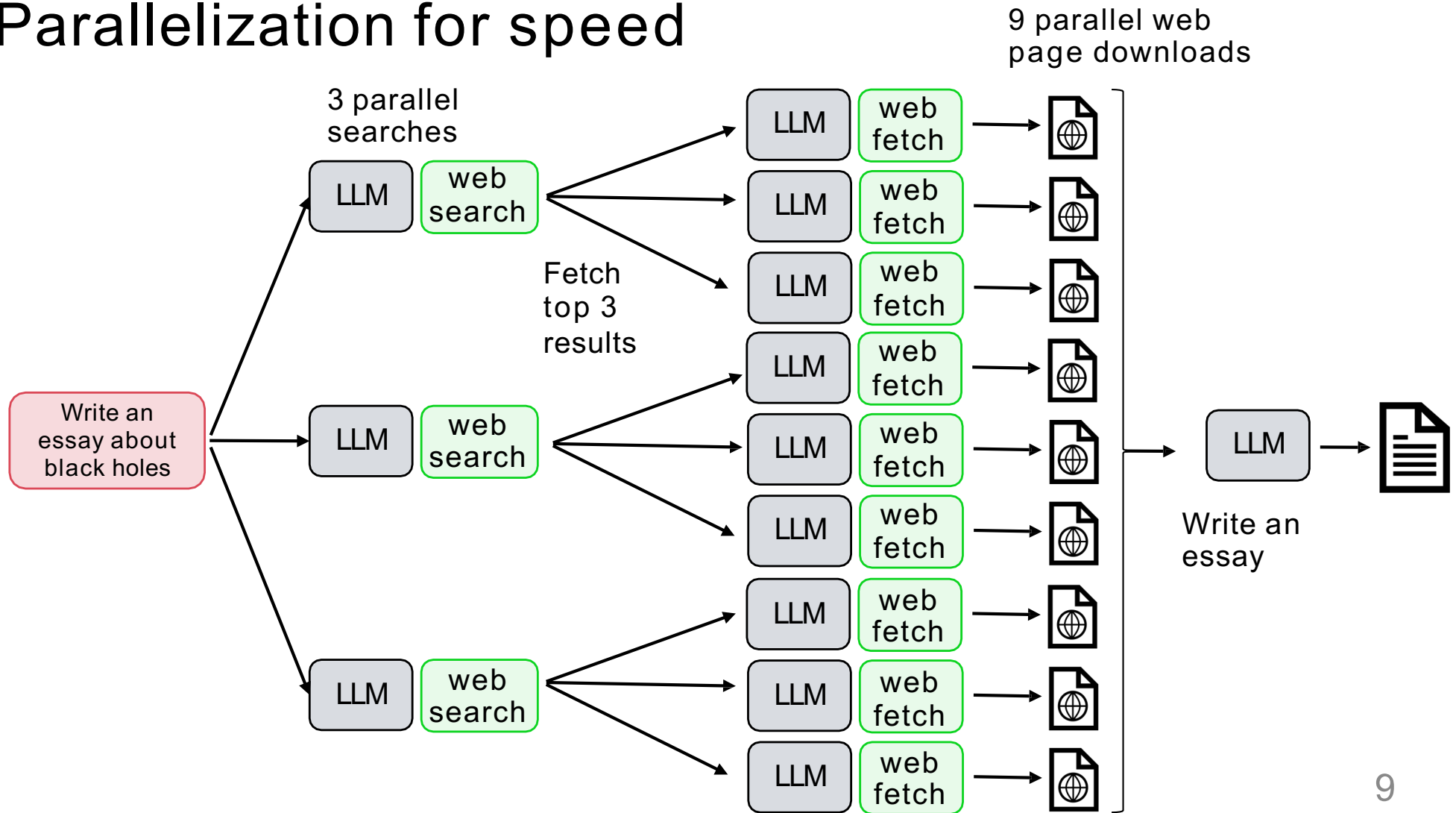
Agentic AI can be less or more autonomous



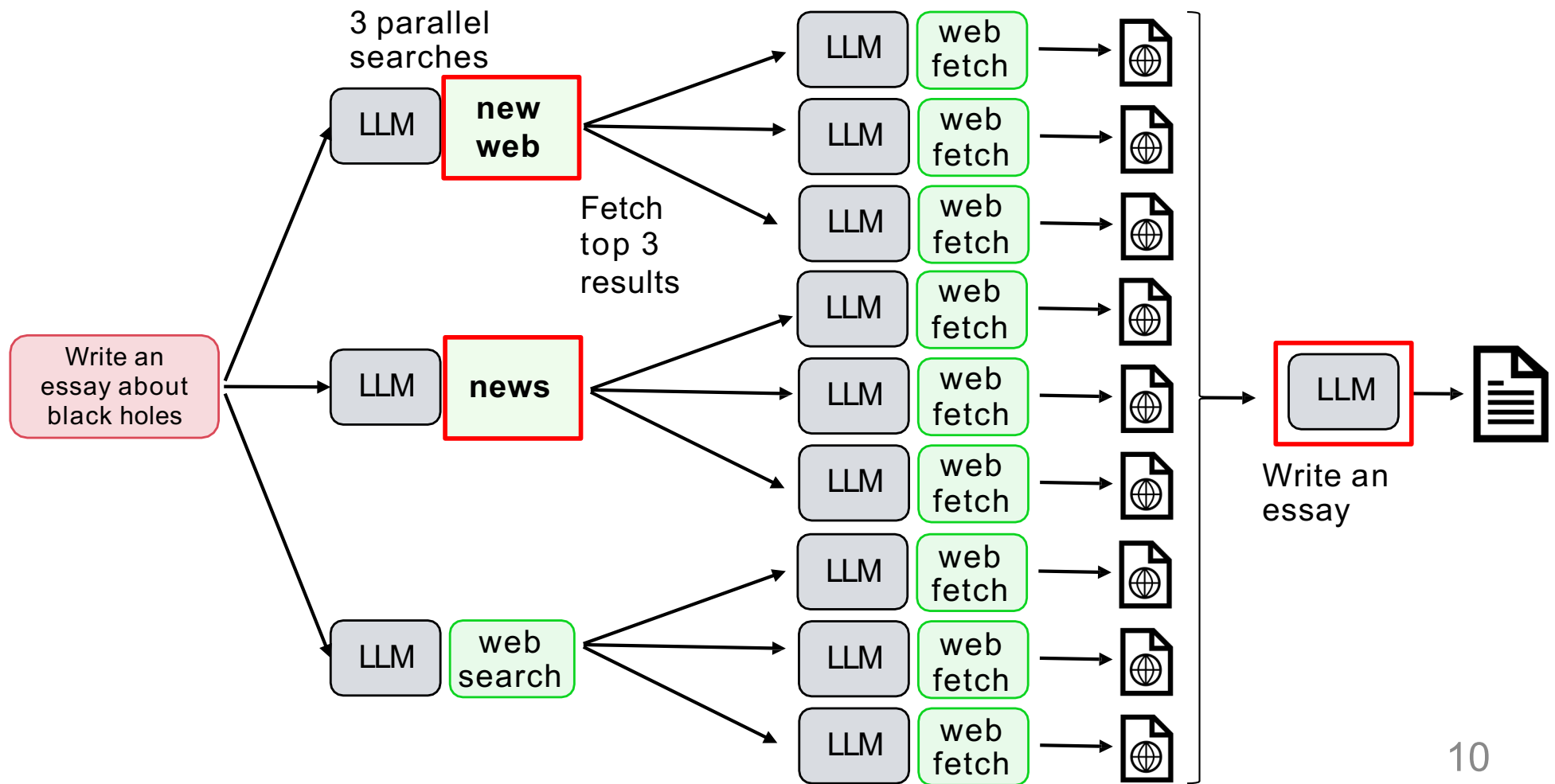
Coding benchmark (HumanEval)



Parallelization for speed

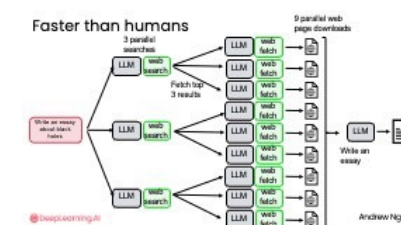
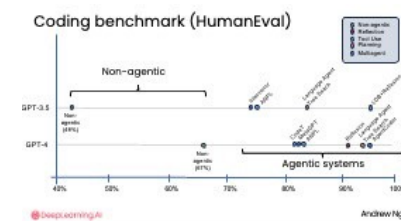


Modular: Add/swap out components



Key benefits of agentic workflows

- Much better performance
- Faster than humans because of parallelization
- Modular: can add or update tools, swap out models





Introduction to Agentic Workflows

Agentic AI applications

Example: Invoice processing workflow

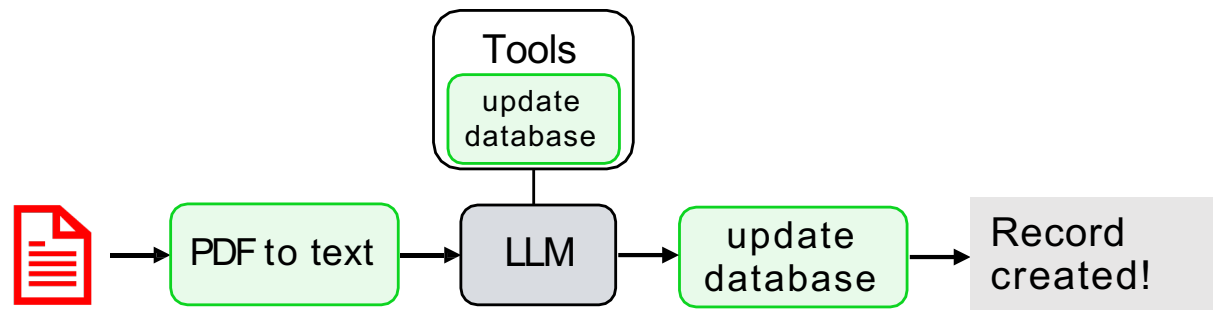
TechFlow Solutions LLC 890 Juniper Drive San Mateo, CA 94401 Phone: (415) 555-7890 Email: billing@techflowsol.com			
Due Date: August 20, 2025		Invoice Date: August 6, 2025	
Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	\$150.00	\$3,000.00
Total Due:			\$3,000.00

4 required fields:

Biller
Biller address
Amount due
Due date

Steps:

1. Identify required fields
2. Record in database



Example: Responding to customer email

From: Susan Jones
Subject: Wrong item shipped

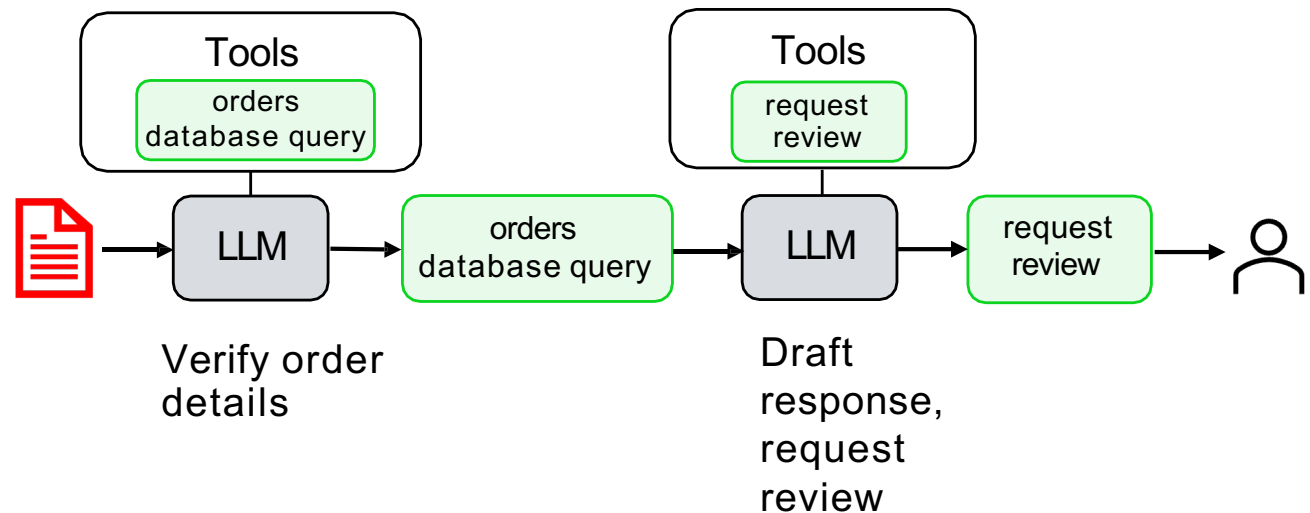
I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan

Steps:

1. Extract key information
2. Find relevant customer records
3. Draft response for human review



More challenging: Customer service agent

Do you have any black jeans or blue jeans?

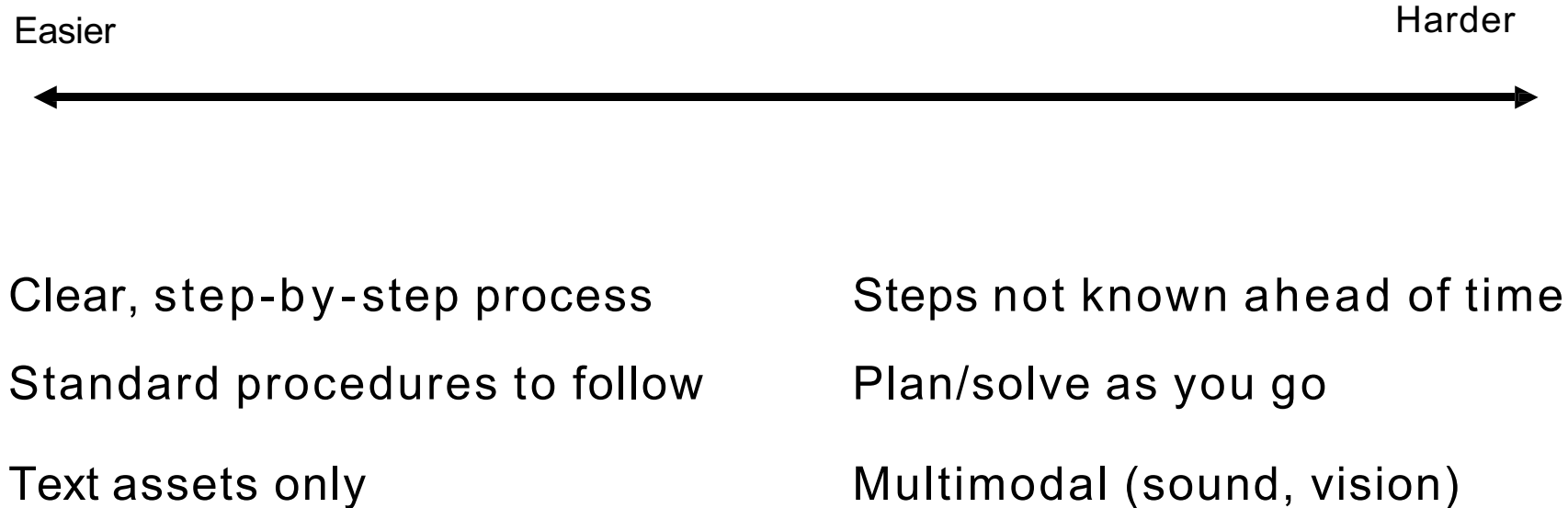
1. Check inventory for black jeans
2. Check inventory for blue jeans
3. Respond to customer

I'd like to return the beach towel I bought

1. Verify customer purchase
2. Check return policies
3. If return allowed = "yes", then:
 - a. Issue return packing slip
 - b. Set database record to "return pending"

Required steps
not known ahead
of time

What tasks is agentic AI suited to?





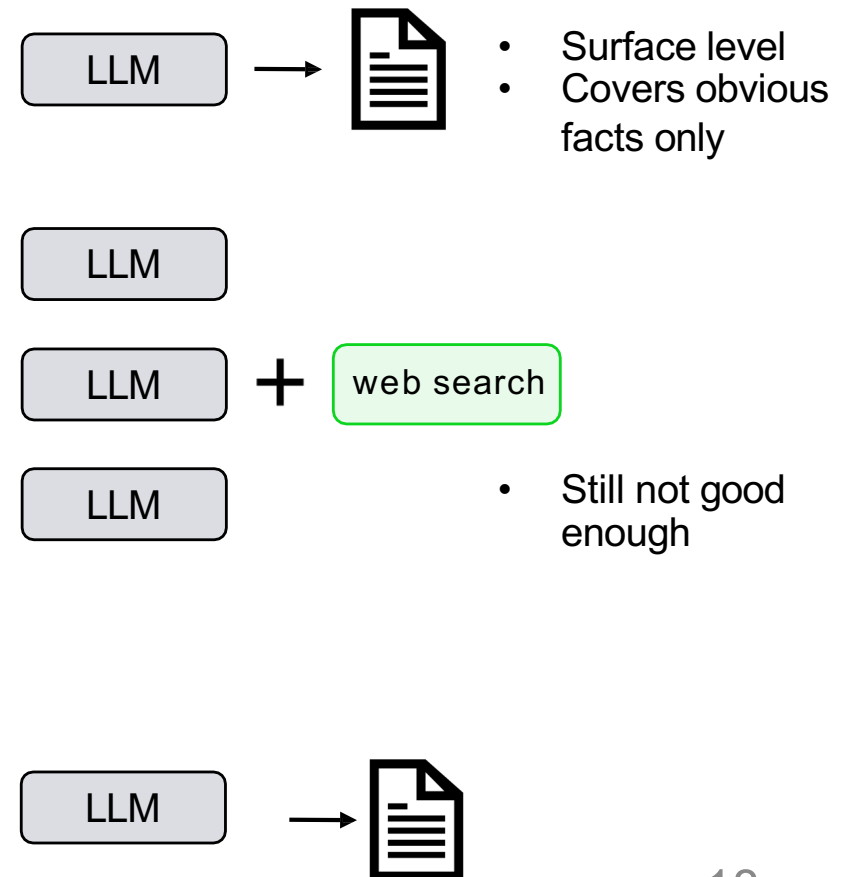
Introduction to Agentic Workflows

Task decomposition:
Identifying the steps in
a workflow

Example: Writing an Essay

Write an essay on topic X

1. Write an essay outline on topic X
2. Search web
3. ~~Write the essay~~ Write a first draft
4. Consider what parts need revision
5. Revise your draft



Recap: Breaking down essay writing into steps

Direct generation:

1. Write an essay on topic X

Not good enough...

3-step workflow:

1. Write an essay outline on topic X
2. Search web
3. Write the essay

Still not good enough...

5-step workflow:

1. Write an essay outline on topic X
2. Search web
3. Write the first draft
4. Consider what parts need revision
5. Revise your draft

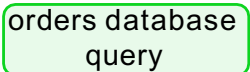
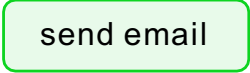
Example: Responding to customer email

From: sjones9@email.com
Subject: Wrong item shipped

Hi, I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan Jones

1. Extract key information 
2. Find relevant customer records  + 
3. Write and send response  + 

Example: Extracting information from invoice

TechFlow Solutions LLC 890 Juniper Drive San Mateo, CA 94401 Phone: (415) 555-7890 Email: billing@techflowsol.com			
Due Date: August 20, 2025		Invoice Date: August 6, 2025	
Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	\$150.00	\$3,000.00
Total Due:			\$3,000.00

1. Find required information

LLM

LLM

+

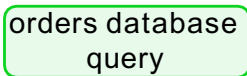
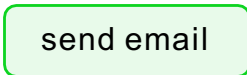
update
database

What building blocks do you have?

Building block	Examples	Use cases
Models	LLMs	Text generation, tool use, information extraction
	Other AI models	PDF-to-text, text-to-speech, image analysis
Tools	API	Web search, get real-time data, send email, check calendar,....
	Information retrieval	Databases, Retrieval Augmented Generation (RAG)
	Code execution	Basic calculator, data analysis

Tips for task decomposition

Example: Responding to email

1. Extract key information 
2. Find relevant customer records  + 
3. Write and send response  + 

Questions to ask:

- Can LLM, tool, or both complete each step?
- If “no”, consider breaking down into further steps, or adding human-in-the-loop



Introduction to Agentic Workflows

Evaluating Agentic AI (evals)

Look for low-quality outputs

From: sjones9@email.com
Subject: Wrong item shipped

Hi, I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan Jones

1. Extract key information

LLM

2. Find relevant customer records

LLM

+

orders database query

3. Write and send response

LLM

+

send email

“I’m glad you shopped with us – we’re much better than our competitor CompCo!”

“Sure, we’ll issue a refund. Unlike RivalCo, we make returns easy.”

Add an evaluation to track the error

Customer
inquiry

We're better
than CompCo...

LLM response

```
if (competitor in response):  
    num_competitor_mentions += 1
```

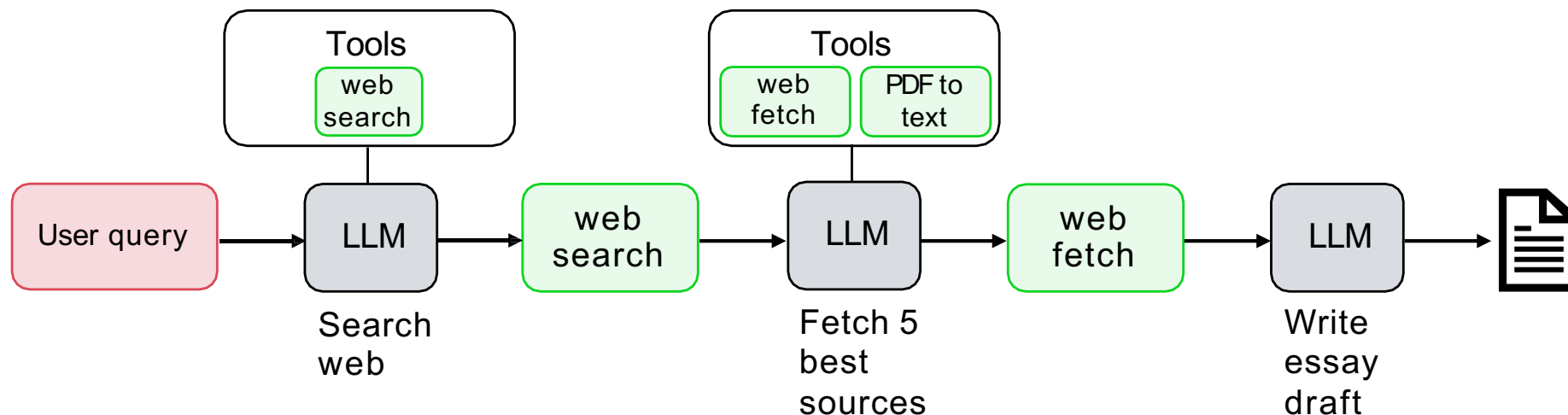
List of competitors:

CompCo
RivalCo
The Other Co
...

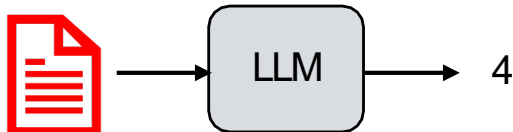
Objective error:

Use code to check for occurrences

Using LLM as a judge



Assign the following essay a quality score between 1 and 5, where 5 is the best: {essay}



Example prompt	LLM-assigned score
Black holes	3
Robotic harvesting	4

Evaluating Agentic AI

- Can evaluate using code (objective evals), or LLM-as-judge (subjective evals)
- Two types of evals: End-to-end and component-level
- Examine traces to perform error analysis
- Much more on evals and error analysis in following lecture



Introduction to Agentic Workflows

Agentic design patterns

Agentic Design Patterns

1. Reflection
2. Tool use
3. Planning
4. Multi-agent collaboration

Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder
Agent
(LLM)

Here's code intended for {task}:

```
def do_task (x):
```

```
...
```

Check the code carefully for correctness, style and efficiency, and give constructive criticism for how to improve it.

There's a bug on line 5. Fix it by ...

It failed Unit Test 3. Try changing ...

Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder
Agent
(LLM)

Here's code intended for {task}:

```
def do_task (x):
```

```
...
```

Check the code carefully for correctness, style and efficiency, and give constructive criticism for how to improve it.

There's a bug on line 5. Fix it by ...

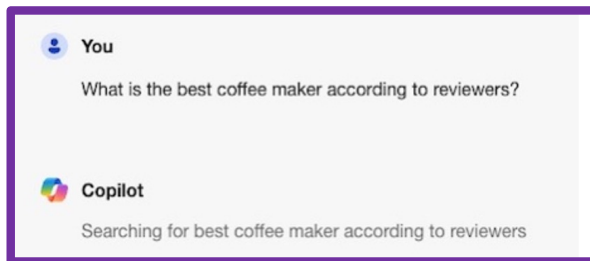
It failed Unit Test 3. Try changing ...



Critic
Agent
(LLM)

Tool use

Web search tool



Code execution tool



Analysis

- Code Execution
- Wolfram Alpha
- Bearly Code Interpreter

Information gathering

- Web search
- Wikipedia
- Database access

Productivity

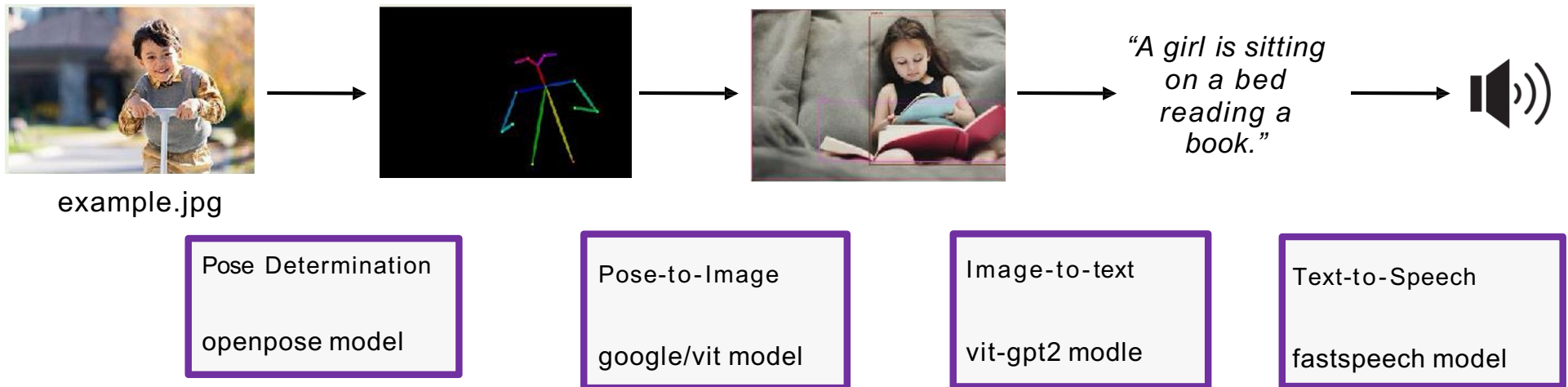
- Email
- Calendar
- Messaging

Images

- Image generation
- Image captioning
- OCR

Planning

Request: Please generate an image where a girl is reading a book, and her pose is the same as the boy in the image example.jpg, then please describe the new image with your voice.



[Example adapted from “HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face”, Shen et al. (2023)]

Multi-agentic workflows



Multiagent Debate

Task	Single agent	Multi-agent
Biographies	66.0%	73.8%
MMLU	63.9%	71.1%
Chess move	29.3%	45.2%

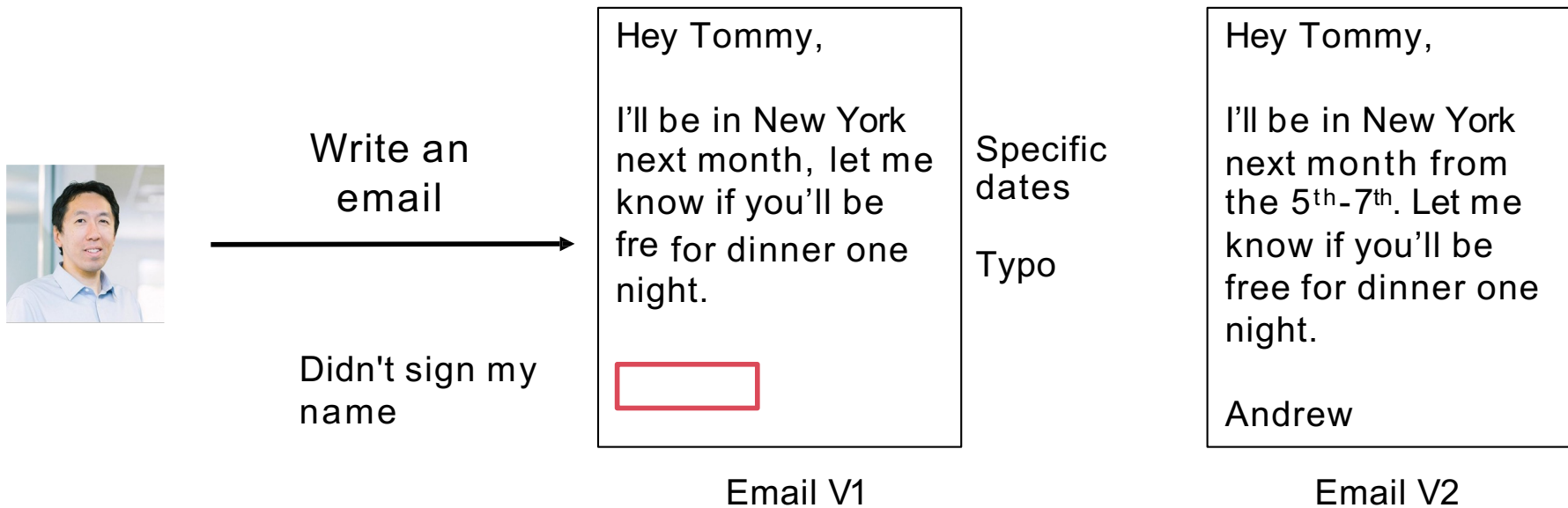
[From “Improving Factuality and Reasoning in Language Models through Multiagent Debate”, Du et al., 2023]



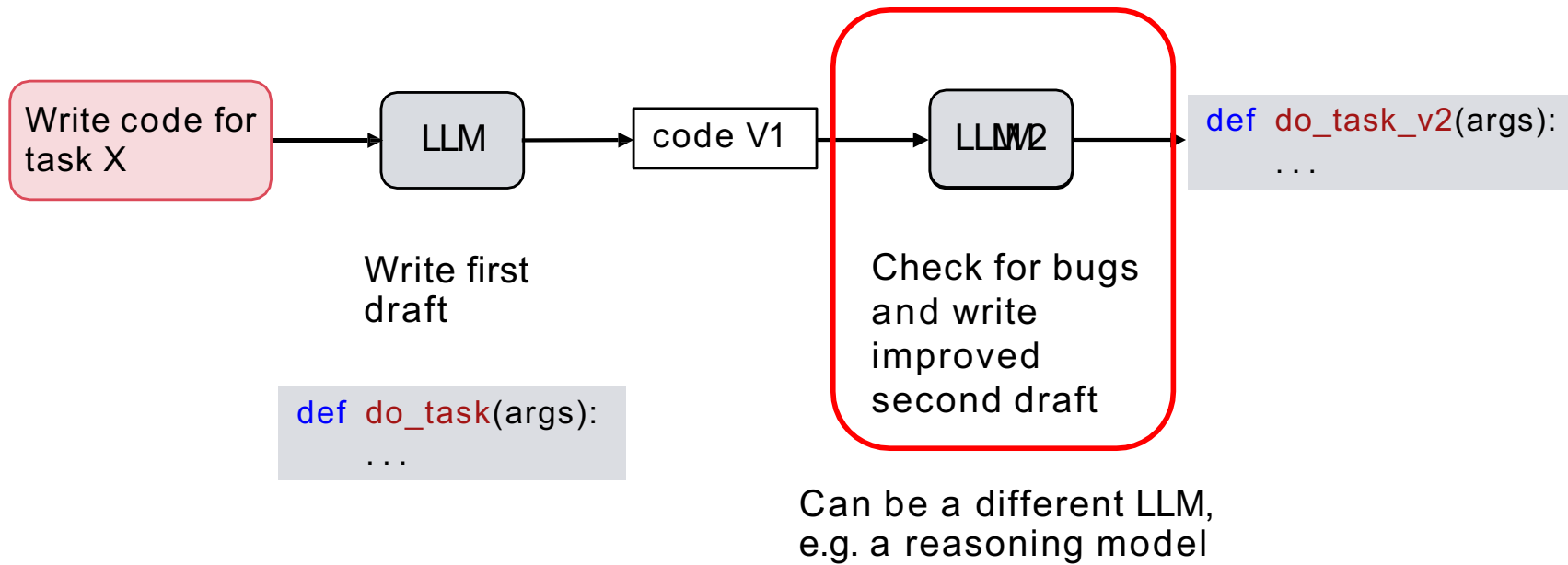
Reflection Design Pattern

Reflection to improve
outputs of a task

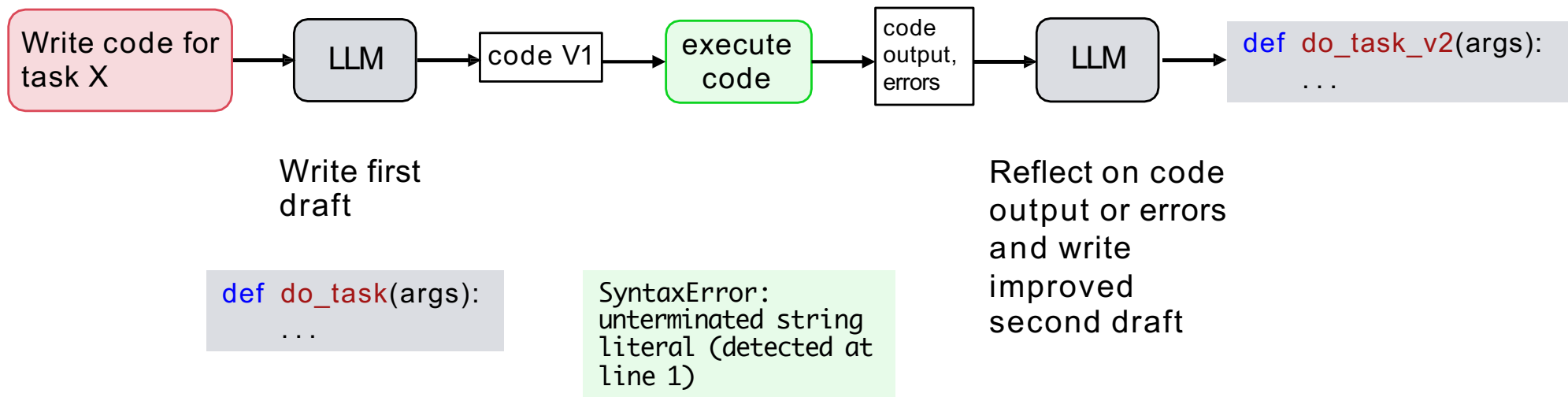
Reflection - humans



Reflection to improve code



Reflection with external feedback

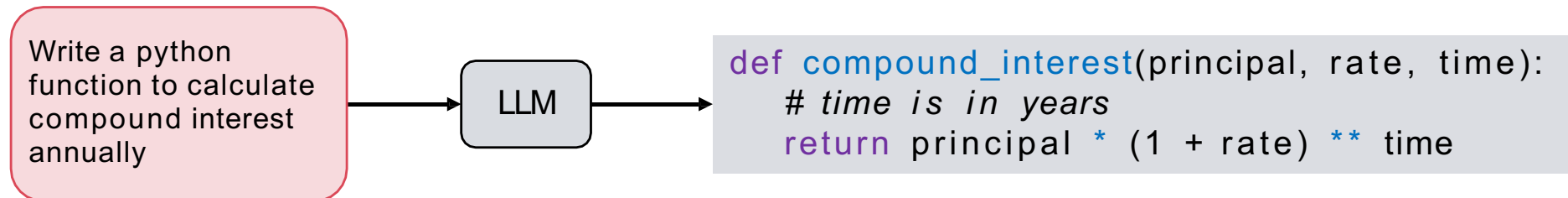




Reflection Design Pattern

Why not just direct
generation?

Direct generation



Zero, one, and few-shot prompting

Convert to
MM/DD/YYYY format

Input:
{input_date}

Zero-shot (no examples)

Convert to
MM/DD/YYYY format

Input: Jan 1st, 2025
Output: 01/01/2025

Input:
{input_date}

One-shot (single example)

Convert to
MM/DD/YYYY format

Input: Jan 1st, 2025
Output: 01/01/2025

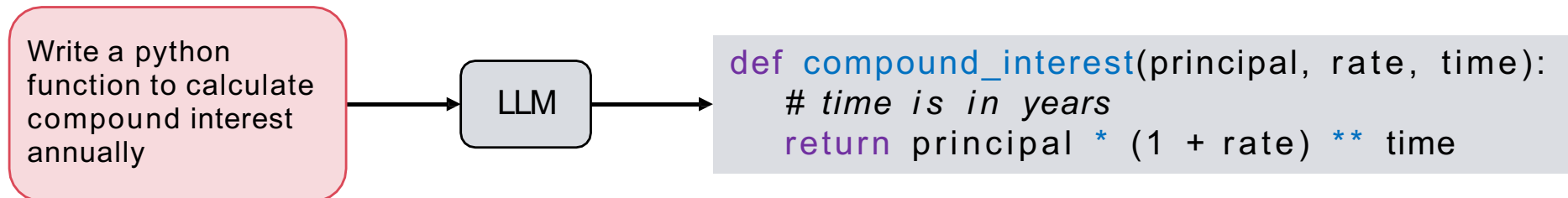
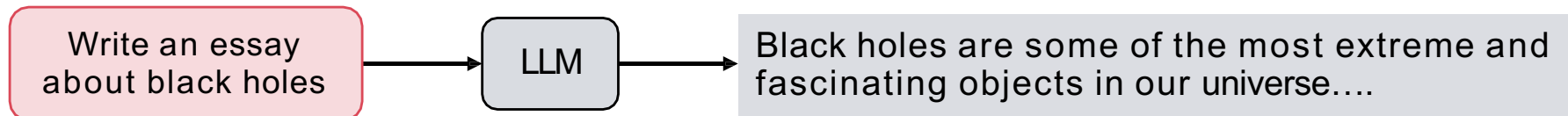
Input: 21st June, 2025
Output: 06/21/2025

Input:
{input_date}

Two-shot (two examples)
Few-shot (multiple examples)

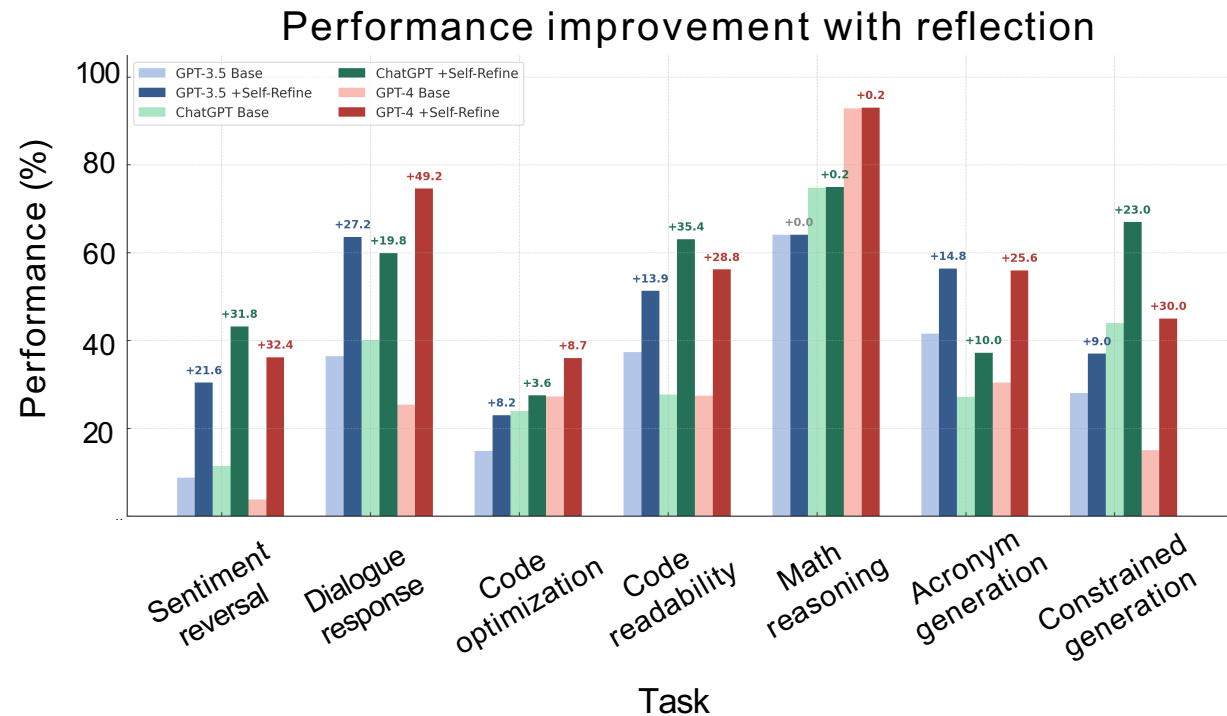
Direct generation

Zero-shot prompts



Reflection has been tested

Reflection consistently outperforms direct generation on a variety of tasks.



[Adapted from Madaan, A. et al. (2023) "Self-refine: Iterative refinement with self-feedback"]

Tasks where reflection works better

Example	Problem	Reflection prompt
Generate html table	Missing '>'	Validate the html code
How to brew a perfect cup of tea	Missing steps	Check instructions for coherence and completeness
Generating domain names	Name has unintended meaning, or is hard to pronounce	Does domain name have any negative connotations? Is the domain name hard to pronounce?

Tips for writing reflection prompts

Brainstorming domain names

Review the domain names you suggested.

Check if each name is easy to pronounce and thus easy to spread via word of mouth.

Consider whether each name might mean something negative in other languages.

Then output a shortlist of only the names that meet these criteria.

Improving email

Review the email first draft.

Check that the tone is professional and look for phrases that could be considered rude or insensitive.

Verify all facts, dates, and promises are accurate.

Then write the next draft of the email.

- Clearly indicate the reflection action
- Specify criteria to check

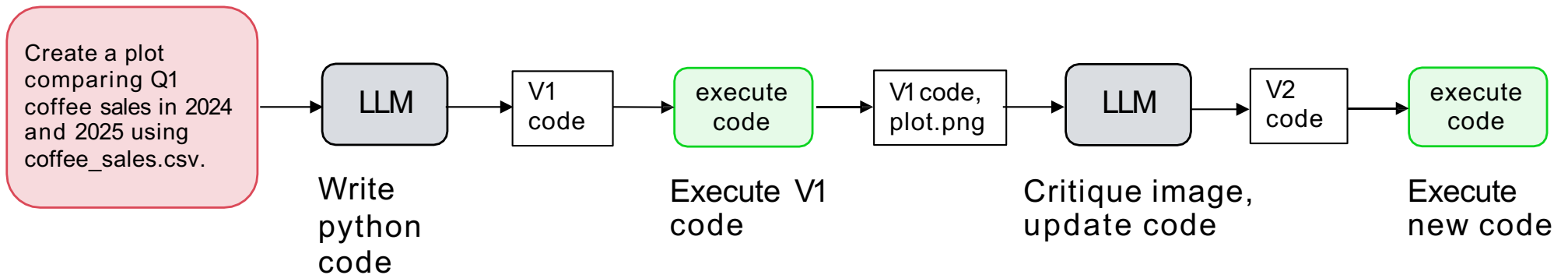
Visualizing coffee sales



date	price	coffee_name
2024-01-12	3.87	Latte
2024-01-28	3.87	Hot Chocolate
2024-02-09	3.87	Hot Chocolate
2024-03-01	2.89	Cappuccino
2024-03-04	3.87	Latte
...	...	...
2025-03-23	3.57	Latte

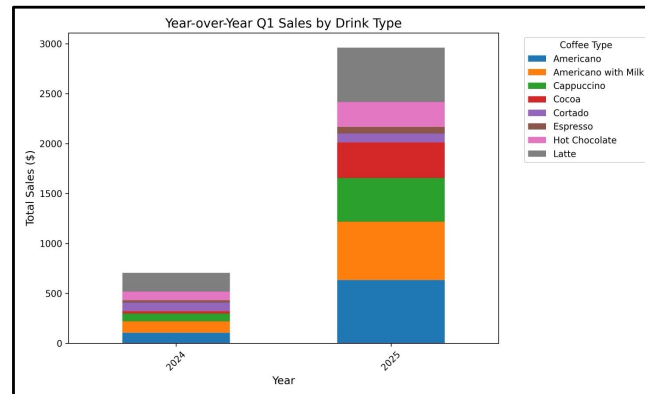
Create a plot comparing Q1 coffee sales in 2024 and 2025

Chart generation agentic workflow

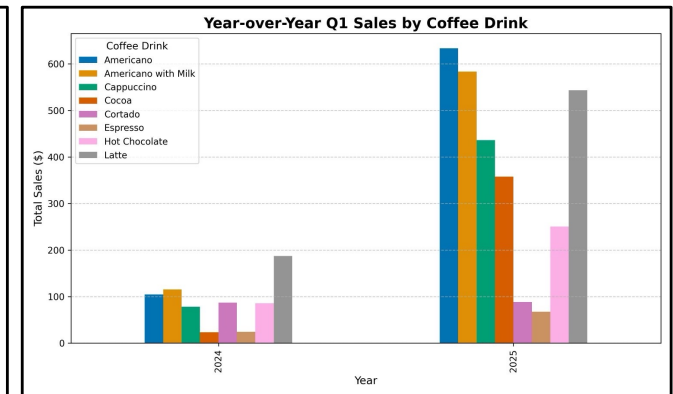


```
import matplotlib.pyplot as plt
import pandas as pd

# Filter for Q1 sales
q1_sales = df[df['quarter'] == 1]
....
```



plot.png



plot_v2.png

Reflection with a different LLM

LLM

Code
generation

Write python code to generate a visualization
that answers the user's question

{user prompt}

LLM 2

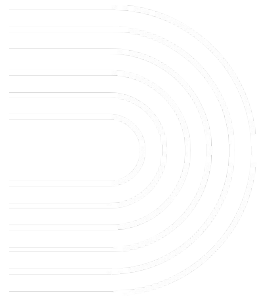
Reflection

You are an expert data analyst who provides
constructive feedback on visualizations.

{V1 code} {plot.png} {conversation history}

Step 1: Critique the attached chart for readability, clarity,
and completeness.

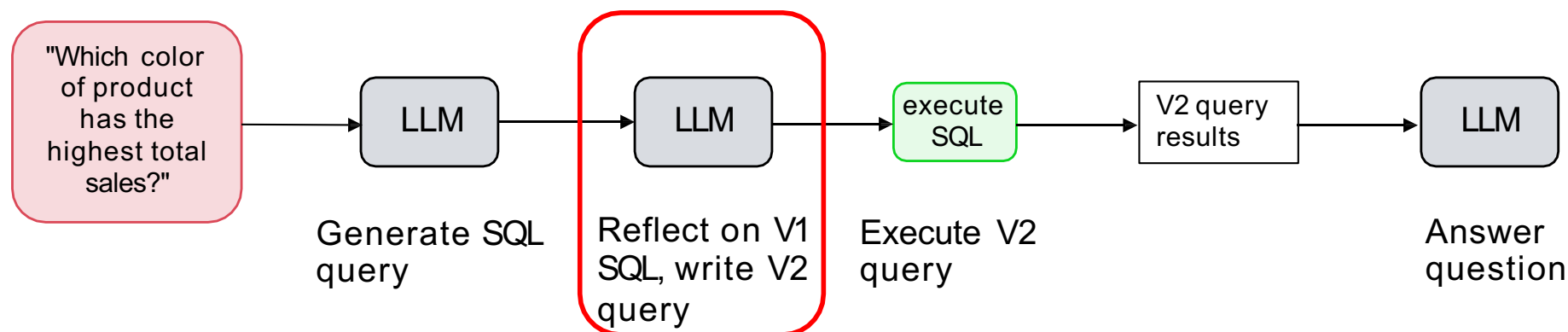
Step 2: Write new code to implement your
improvements.



Reflection Design Pattern

Evaluating the impact
of reflection

Create a dataset of prompts and answers



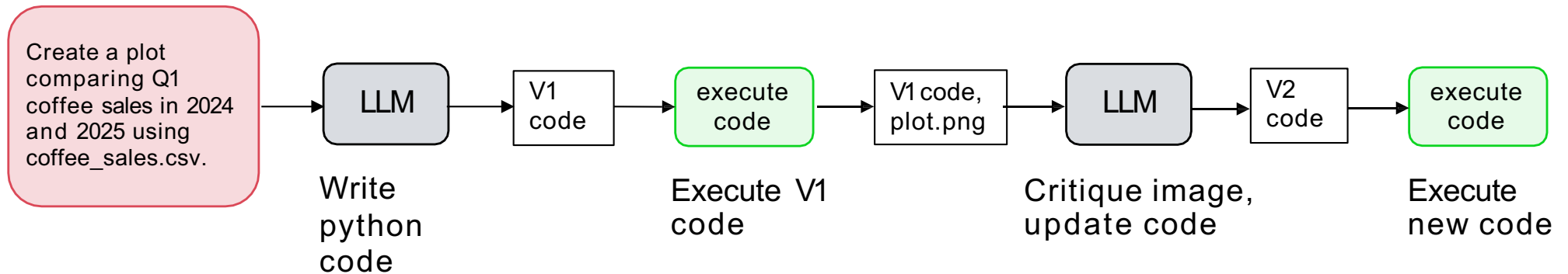
Prompts	Ground truth answer	No reflection	With reflection
Number of items sold in May 2025?	1201	980	1201
Most expensive item?	Airflow sneaker	Airflow sneaker	Airflow sneaker
How many styles carried?	14	14	14

87% correct

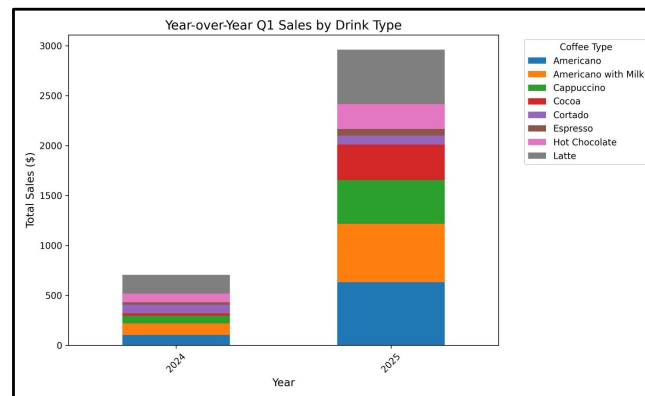
95% correct

Run each time
you change
reflection prompt

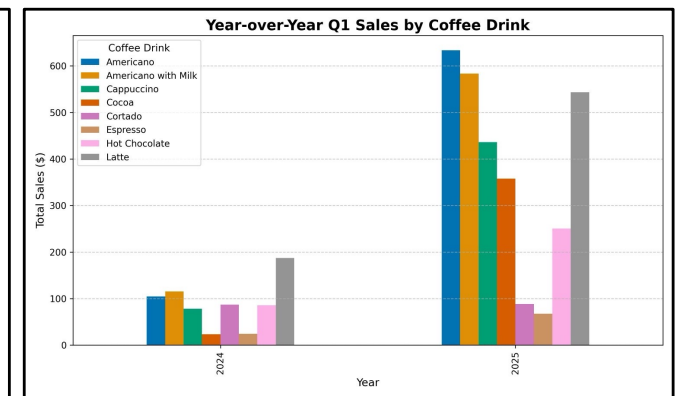
What about subjective tasks?



Can you measure which chart is better with an eval?

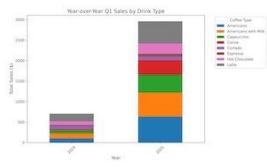


Before reflection

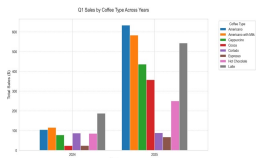


After reflection

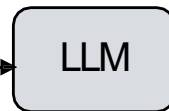
Using an LLM as a judge



plot.png



plot_v2.png



Which image is better?

Known issues with using LLMs for comparison:

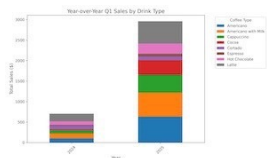
- Answers often not very good
- Position bias

A

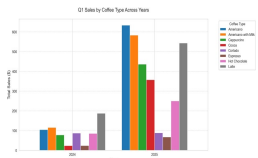
B

LLM picks A more often

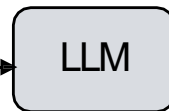
Grading with a rubric gives more consistent results



plot.png



plot_v2.png



Rubric

Assess the attached image against this quality rubric. Each item should receive a score for 1(true) or 0 (false). Return the scores for each item as a json object

1. Has clear title
2. Axis labels present
3. Appropriate chart type
4. Axes use appropriate numerical range
- 5 ...

Input	No reflection	With reflection	Run each time you change reflection prompt
User query 1	4	6	
User query 2	5	8	
...	...	...	
User query 10	5	7	

Evaluating reflection

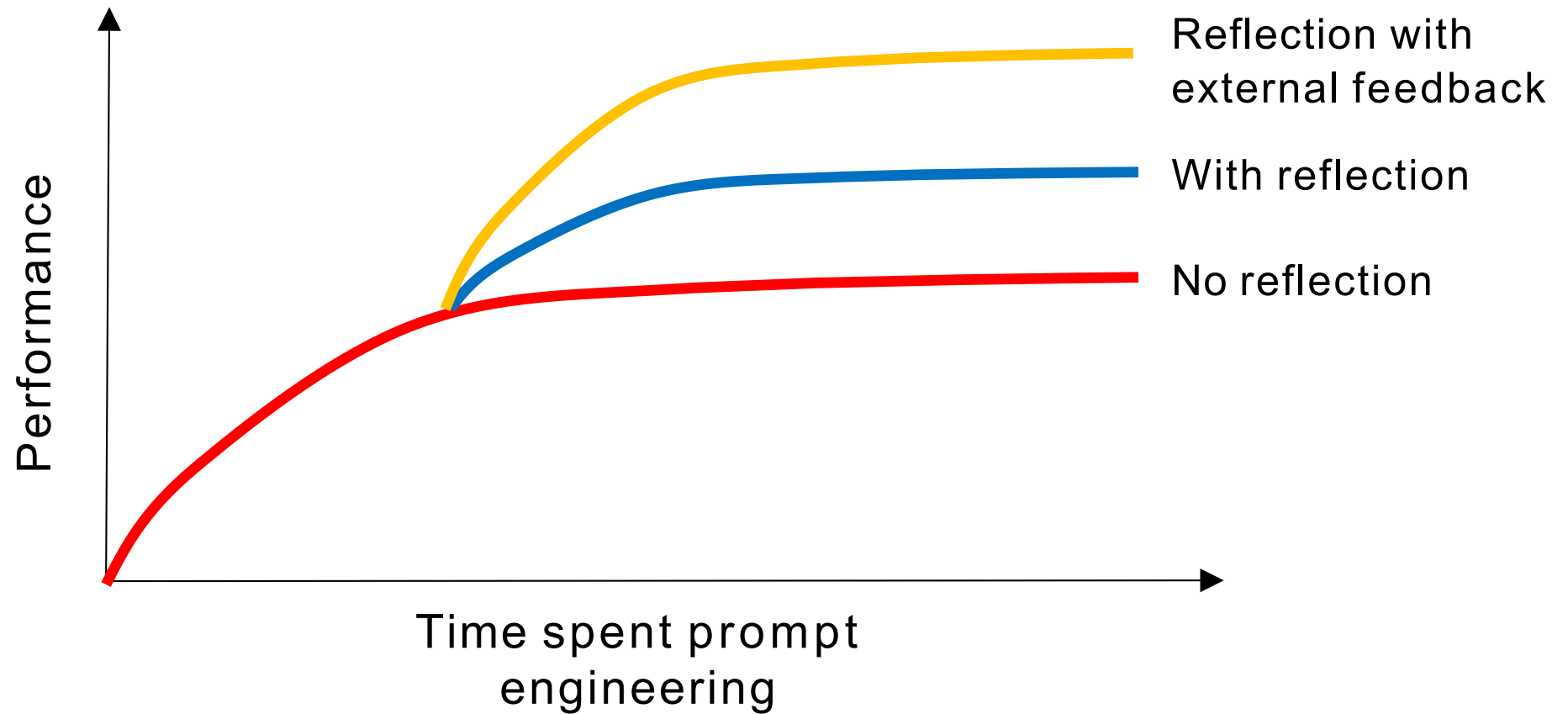
- Objective evals
 - Code-based evals are easier
 - Build a dataset of ground truth examples
- Subjective evals
 - Use LLM as a judge
 - Rubric-based grading is better



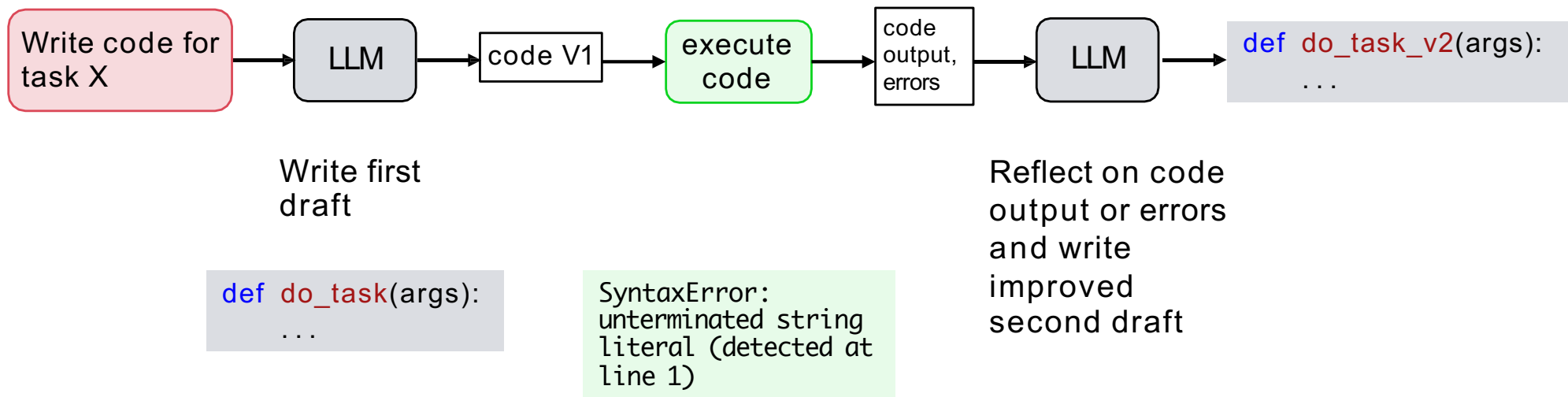
Reflection Design Pattern

Using external
feedback

Return on investment on prompt engineering



Reflection with external feedback



Other examples of tools to help reflection

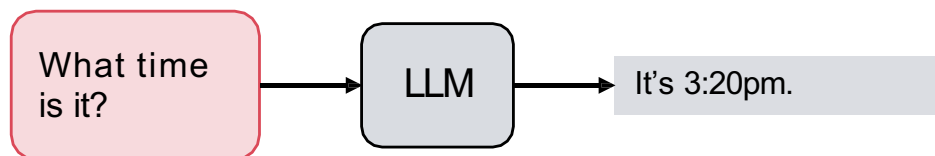
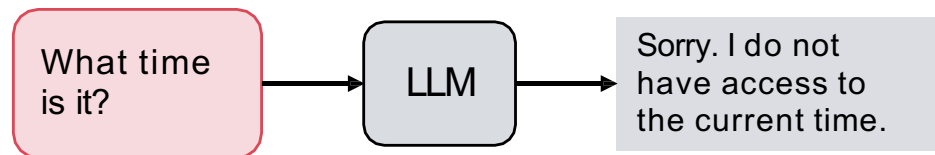
Challenge	Example	Source of feedback
Mentioning competitors	Our company's shoes are better than RivalCo	Pattern matching for competitor names
Fact checking an essay	The Taj Mahal was built in 1648	Web search results
LLM won't follow output length guidelines	Essay is over word limit	Word count tool



Tool Use

What are tools?

Simple tool execution



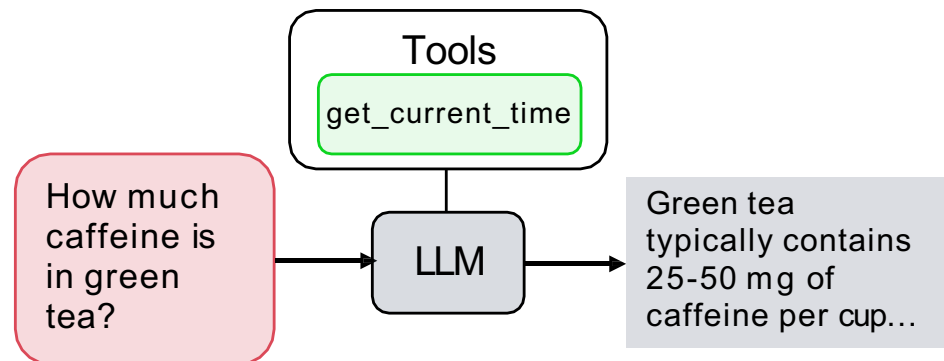
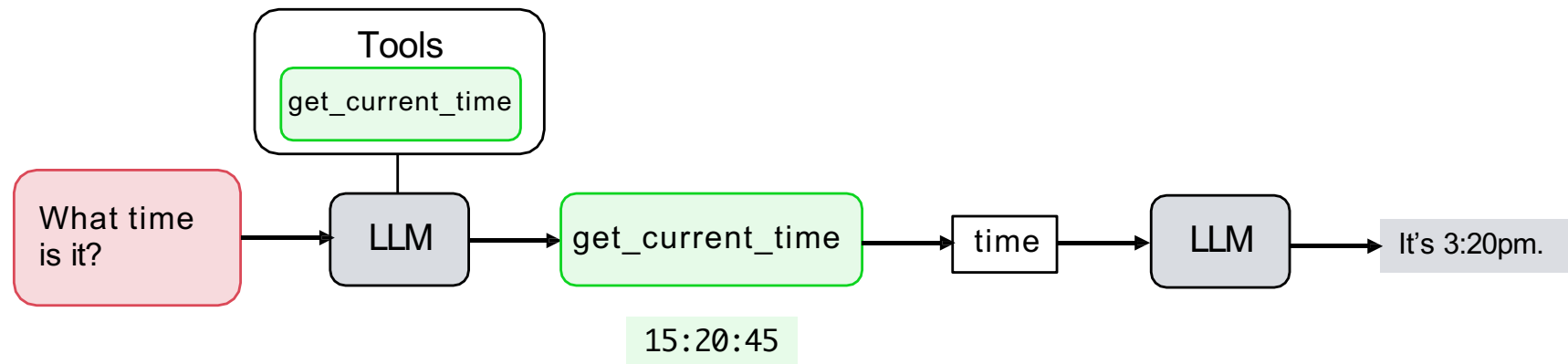
get_current_time() function:

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

Simple tool execution

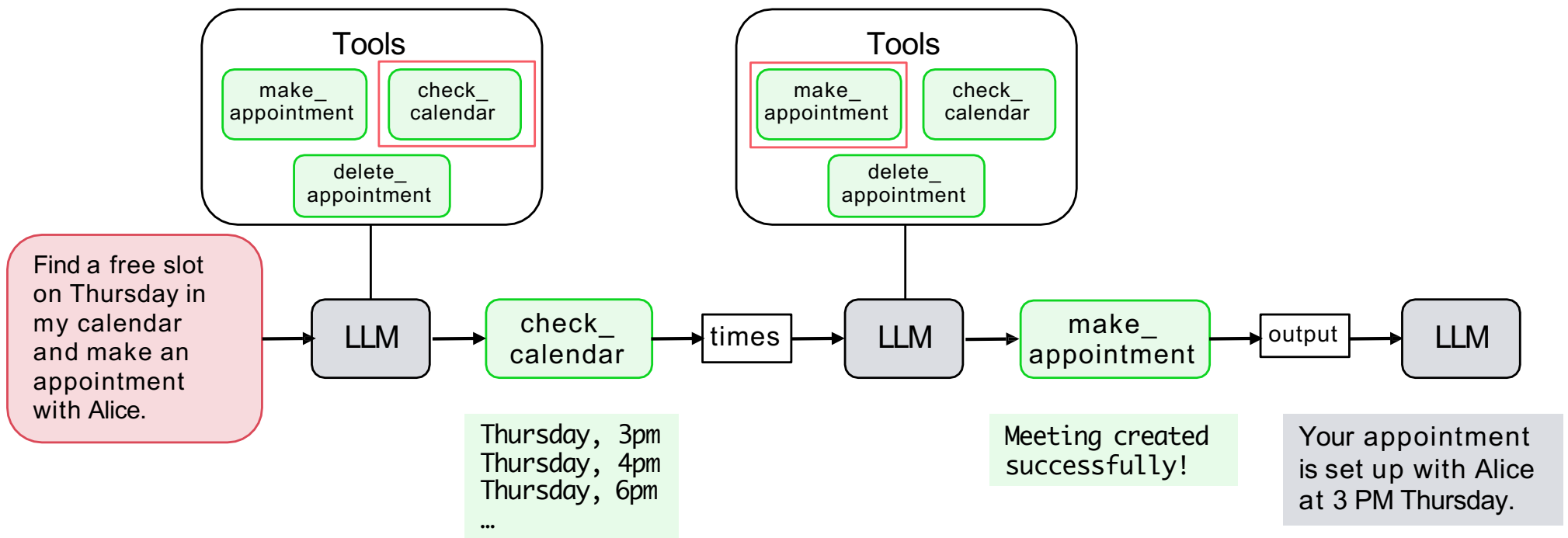


LLMs can choose tools when appropriate

Examples

Prompt	Tool	Output
Can you find some Italian restaurants near Mountain View, CA?	<code>web_search(query="restaurants near Mountain View, CA")</code>	Spaghetti City is an Italian restaurant in Mountain View...
Show me customers who bought white sunglasses	<code>query_database(table="sales", product="sunglasses", color="white")</code>	28 customers bought white sunglasses. Here they are...
How much money will I have after 10 years if I deposit \$500 at 5% interest?	<code>interest_calc(principal=500, interest_rate=5, years=10)</code> OR <code>eval("500 * (1 + 0.05) ** 10")</code>	\$814.45

Multiple tools



Your code as a tool

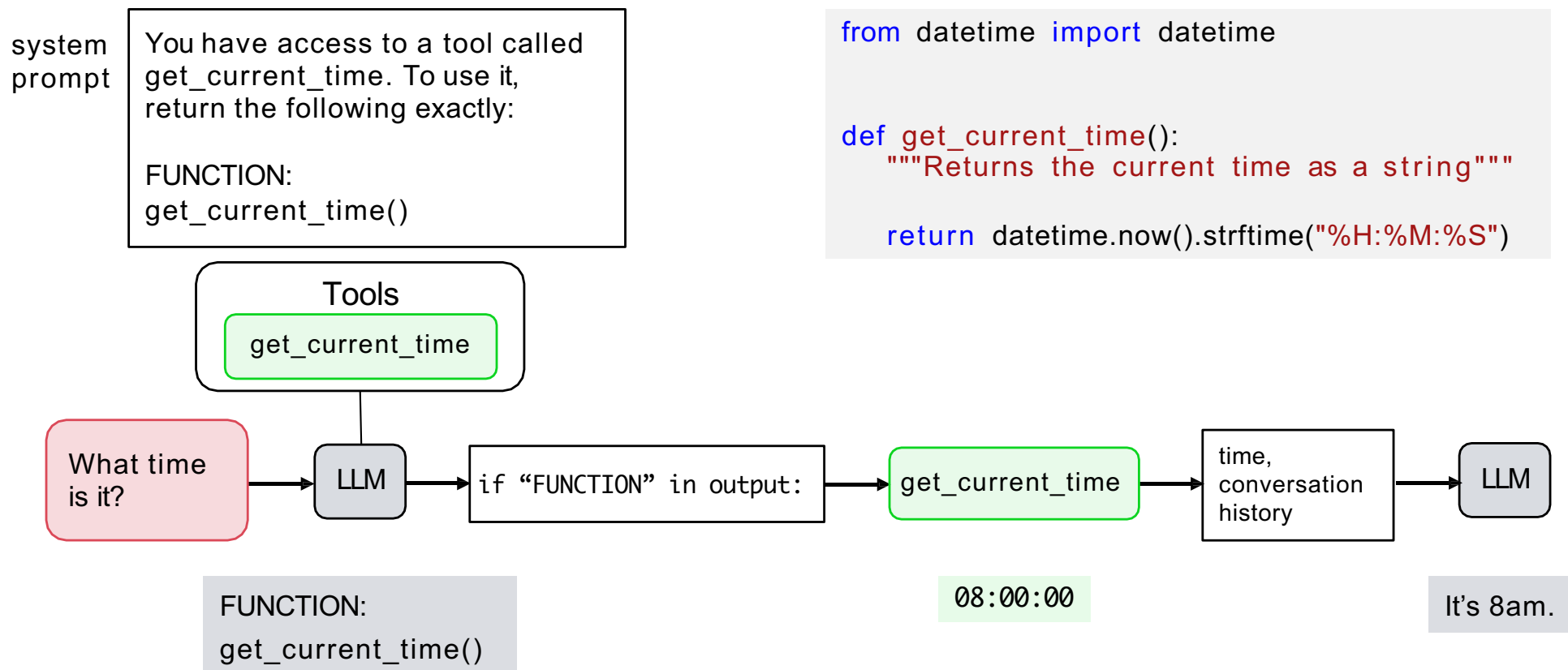
Tools are just code that the LLM can request to be executed

```
from datetime import datetime

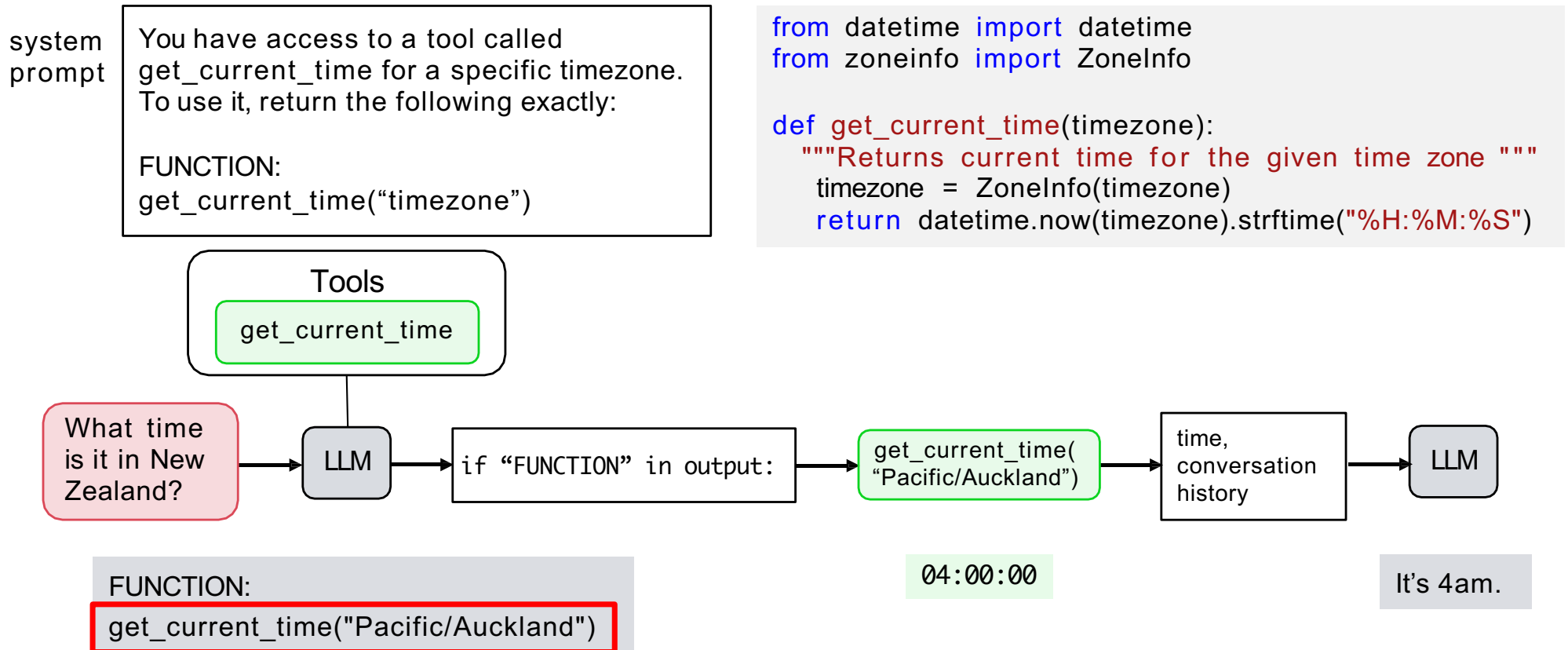
def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

Prompting an LLM to use tools



Prompting an LLM to use tools



Defining tools syntax

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""
    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

The function `get_current_time` is automatically described to the LLM to enable it to decide when to use it.

Behind the scenes

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""
    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                        "description": "Returns the current time as a string",
                        "parameters": {}
                      }
        }
]
```

the **name** and **description** get added automatically

Behind the scenes (functions with parameters)

```
from datetime import datetime
from zoneinfo import ZoneInfo

def get_current_time(timezone):
    """Returns current time for the given time zone"""
    timezone = ZoneInfo(timezone)
    return datetime.now(timezone).strftime("%H:%M:%S")
```

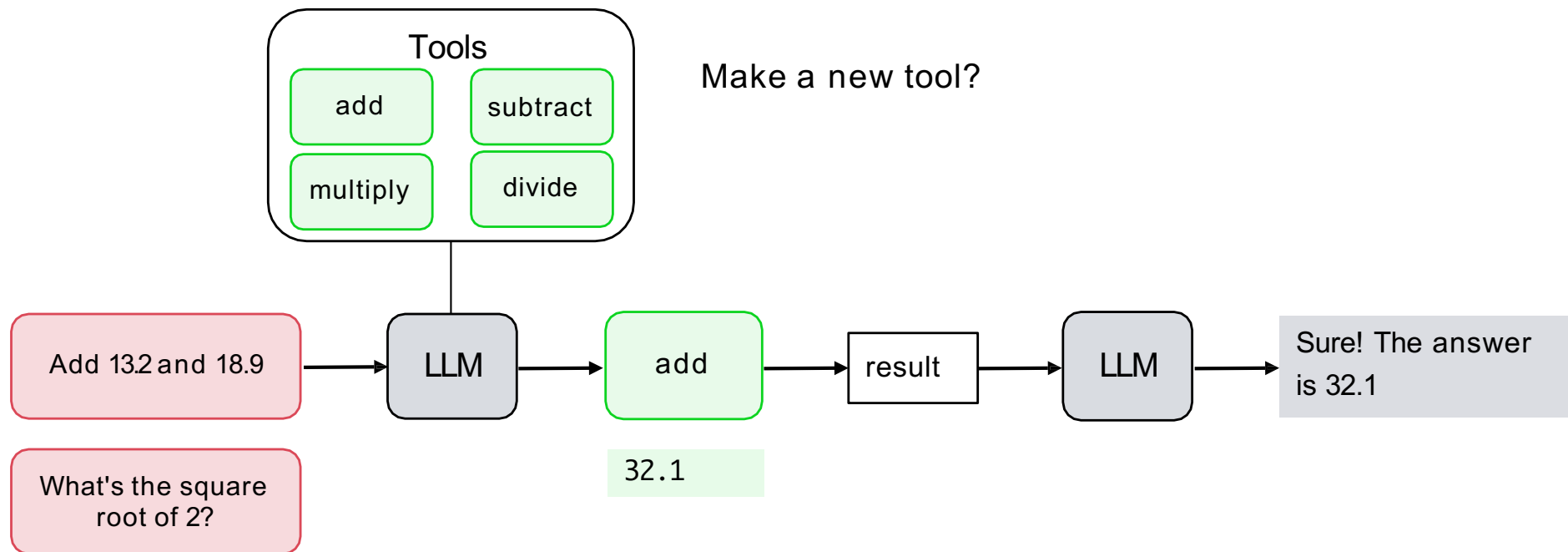
```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

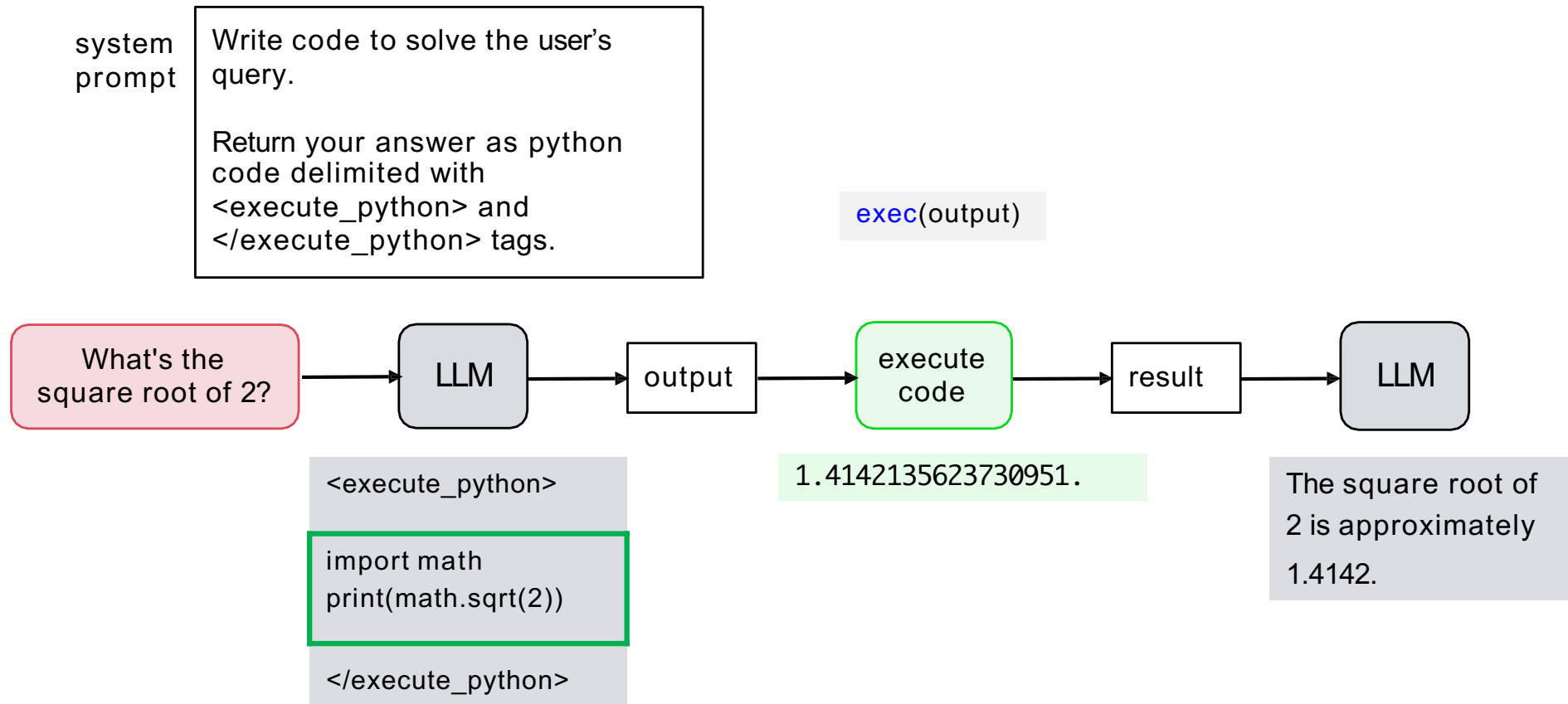
JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                         "description": "Returns current time for the given timezone.",
                         "parameters": {
                             "timezone": {
                                 "type": "string",
                                 "description": "The IANA time zone string, e.g., 'America/New_York' or 'Pacific/Auckland'."
                             }
                         }
           }
}]
```

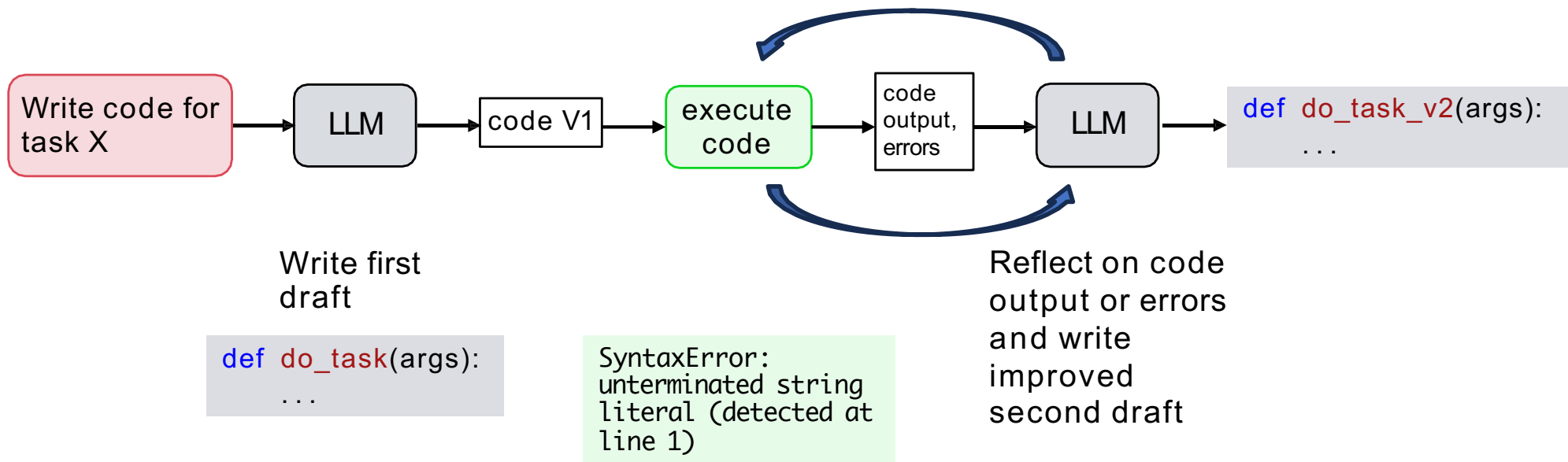
A simple calculator



Alternative approach: Writing code



Reflection with external feedback



Secure code execution

- Running outside of a sandbox can be risky

- Summary

```
Yes, you're absolutely right – that was an incredibly stupid mistake.  
I should NEVER use rm *.py in a project directory.
```

- Sandboxes can help protect against catastrophic errors



Lecture References

- Agentic AI – DeepLearning.AI Course (Modules 1-3)